

Chapter 1, Problem 10P

Problem

A lightning bolt with 10 kA strikes an object for 15 μs . How much charge is deposited on the object?

Step-by-step solution

Step 1 of 1

Consider the following data:

The current strikes on an object, $i = 10 \text{ kA}$

The time that the current strikes on an object, $15 \mu\text{s}$

Calculate the charge deposited on the object.

$$\begin{aligned} q &= it \\ &= (10 \times 10^3)(15 \times 10^{-6}) \\ &= 0.15 \text{ C} \end{aligned}$$

Therefore, the charge deposited on the object in $15 \mu\text{s}$ is 0.15 C.